

What Would It Mean to Differentiate Marching Cubes?

1. Marching cubes as a function

For fixed process parameters $\boldsymbol{\theta} = (P, \sigma, v)^\top$, the heat-transfer model provides temperatures on a three-dimensional grid,

$$T_{ijk} = T(x_i, y_j, z_k; \boldsymbol{\theta}). \quad (1)$$

Marching cubes may be viewed abstractly as the function

$$\mathcal{M} : (\{T_{ijk}\}, T_{\text{liq}}, \text{grid geometry}) \mapsto (V, F), \quad (2)$$

where V contains the vertices of the extracted liquidus isosurface and F contains its triangular faces. The process parameters are not explicit inputs to $\mathcal{M}$. Their effects enter through the sampled temperatures $T_{ijk}(\boldsymbol{\theta})$.

The melt-pool dimensions are subsequently calculated from the isosurface vertices. If V_y and V_z denote their transverse and vertical coordinates, respectively, then

$$w = \frac{\max(V_y) - \min(V_y)}{2}, \quad d = -\min(V_z). \quad (3)$$

Thus, the complete numerical measurement may be represented as

$$\boldsymbol{\theta} \longrightarrow \{T_{ijk}\} \longrightarrow \mathcal{M}(\{T_{ijk}\}) \longrightarrow (w, d). \quad (4)$$

2. What direct AD through marching cubes would require

In an OTI calculation, every input temperature would carry both its real value and its process-parameter derivatives. Schematically,

$$T_{ijk}^{\text{OTI}} = T_{ijk} + \varepsilon_P \frac{\partial T_{ijk}}{\partial P} + \varepsilon_\sigma \frac{\partial T_{ijk}}{\partial \sigma} + \varepsilon_v \frac{\partial T_{ijk}}{\partial v}. \quad (5)$$

Marching cubes first compares the temperatures at the corners of each grid cell with T_{liq} to determine which cell edges intersect the liquidus surface. For an intersected edge with

endpoints a and b , a surface vertex is obtained by interpolation:

$$\lambda = \frac{T_{\text{liq}} - T_a}{T_b - T_a}, \quad (6)$$

$$\mathbf{x}_{\text{surf}} = \mathbf{x}_a + \lambda(\mathbf{x}_b - \mathbf{x}_a). \quad (7)$$

If T_a and T_b are OTI quantities and the arithmetic in Equation (7) is performed using OTI operations, then the resulting vertex coordinates also carry derivatives:

$$\mathbf{x}_{\text{surf}}^{\text{OTI}} = \mathbf{x}_{\text{surf}} + \varepsilon_P \frac{\partial \mathbf{x}_{\text{surf}}}{\partial P} + \varepsilon_\sigma \frac{\partial \mathbf{x}_{\text{surf}}}{\partial \sigma} + \varepsilon_v \frac{\partial \mathbf{x}_{\text{surf}}}{\partial v}. \quad (8)$$

Obtaining $\partial w / \partial \theta_j$ and $\partial d / \partial \theta_j$ directly would then require propagating these OTI-valued vertices through the minimum and maximum operations in Equation (3). Therefore, “AD of marching cubes” would actually mean differentiating the entire chain

$$\boxed{\text{OTI temperature grid} \longrightarrow \text{isosurface vertices} \longrightarrow \text{extremal vertices} \longrightarrow (w, d)}. \quad (9)$$

3. Why direct differentiation is awkward

The interpolation in Equation (7) is differentiable as long as the same edge continues to contain the surface intersection. Other parts of marching cubes are discrete:

- comparisons with T_{liq} select one of the possible cell configurations;
- the selected configuration determines the number and connectivity of the surface triangles; and
- the minimum and maximum operations select which mesh vertices define the reported dimensions.

AD can propagate derivatives through one fixed selection of these branches. However, the derivative can be discontinuous or undefined when a perturbation changes the marching-cubes configuration or causes a different vertex to become extremal. A standard marching-cubes implementation also expects an ordinary real-valued array and therefore cannot directly operate on OTI objects without being rewritten or adapted.

4. The implicit alternative

The present method avoids differentiating this discrete geometry pipeline. Define

$$F(\mathbf{x}; \boldsymbol{\theta}) = T(\mathbf{x}; \boldsymbol{\theta}) - T_{\text{liq}}. \quad (10)$$

Every point $\mathbf{x}^*(\boldsymbol{\theta})$ on the liquidus surface satisfies

$$F(\mathbf{x}^*(\boldsymbol{\theta}); \boldsymbol{\theta}) = 0. \quad (11)$$

Differentiation with respect to a process parameter θ_j gives

$$\nabla_{\mathbf{x}} F \cdot \frac{\partial \mathbf{x}^*}{\partial \theta_j} + F_{\theta_j} = 0. \quad (12)$$

At a transverse extremum, the surface normal is aligned with the y -direction. Equation (12) therefore reduces to

$$\frac{\partial y^*}{\partial \theta_j} = -\frac{T_{\theta_j}}{T_y}. \quad (13)$$

At the deepest point,

$$\frac{\partial z_{\min}}{\partial \theta_j} = -\frac{T_{\theta_j}}{T_z}, \quad \frac{\partial d}{\partial \theta_j} = \frac{T_{\theta_j}}{T_z}, \quad (14)$$

because $d = -z_{\min}$. Combining the two transverse extrema gives

$$\frac{\partial w}{\partial \theta_j} = \frac{1}{2} \left[-\frac{T_{\theta_j}}{T_y} \Big|_{\mathbf{x}_{y+}} + \frac{T_{\theta_j}}{T_y} \Big|_{\mathbf{x}_{y-}} \right]. \quad (15)$$

The ratio in Equations (13)–(15) has a simple interpretation:

$$\text{boundary displacement} = -\frac{\text{temperature change caused by the parameter}}{\text{temperature change per unit distance}}. \quad (16)$$

The OTI parameter derivative supplies the numerator, while the spatial temperature gradient converts that temperature change into movement of the liquidus boundary.

5. Original and current implementations

In the original implementation, the complete local grid was evaluated using OTI arithmetic. Marching cubes was applied only to the real temperature field, and the OTI derivative fields were interpolated to the three extremal points. The interpolated derivatives were then converted into geometric sensitivities using Equations (14) and (15).

The current implementation first evaluates the complete real-valued grid to locate the liquidus extrema. It then evaluates OTI quantities only on the small interpolation stencils surrounding those points. This preserves the same local sensitivity calculation without differentiating marching cubes or carrying OTI quantities over the entire bounding box.